Q12 — Comprehension check

DLS 2026 — AI Internals Module

Student question

"Why do models give confident answers even when they are wrong?"

This comprehension check asks you to connect the surface phenomenon of "confident wrongness" to concrete facts about how language models are trained, calibrated, and decoded.

Question. A large language model answers a factual question fluently and authoritatively, yet the stated fact is false (a "hallucination"). Explain *why* such models tend to express high confidence even when wrong. In your answer, address **all four** of the following mechanisms, and state briefly how each contributes to the mismatch between expressed confidence and actual correctness:

(a) the training objective (next-token likelihood) and what it does *not* reward;

(b) *miscalibration* introduced or reshaped by post-training (RLHF / instruction tuning);

(c) the effect of *decoding* and token-level probabilities on the fluency of a wrong answer;

(d) why the model has no reliable internal signal that cleanly separates "I know" from "I am guessing."

Answer. Confident wrongness is not a single bug but the joint product of the objective, the alignment procedure, and the decoding regime.

(a) The objective rewards plausibility, not truth. Autoregressive models are trained to maximize the likelihood of the next token given the context [1, 10]. This objective rewards producing the *statistically most plausible continuation* of the preceding text, including the fluent, assertive style of confident prose that dominates the training corpus. Nothing in the loss distinguishes a true statement from a false-but-plausible one; a well-formed wrong answer and a well-formed right answer can incur nearly identical per-token loss. Hallucination is therefore partly intrinsic: Kalai and Vempala [4] and related analyses argue that models optimized this way will inevitably generate some fabricated but statistically typical content, and that standard training and evaluation *reward guessing over abstaining* [5].

(b) Post-training reshapes (and often degrades) calibration. A pretrained base model can be surprisingly well-calibrated on multiple-choice tasks: its probabilities roughly track the likelihood of being correct [3]. However, RLHF and instruction tuning optimize for human-preferred responses, and human raters tend to prefer answers that are decisive, helpful-sounding, and confident. OpenAI's GPT-4 report explicitly notes that the base model was well-calibrated but that calibration *decreased* after RLHF post-training [8]. The model thus learns a stylistic association between confidence and reward, decoupling expressed certainty from epistemic warrant. Preference optimization can also induce *sycophancy*—agreeing with or flattering the user—which further rewards assured-sounding but unjustified claims [9].

(c) Decoding produces fluent surface form regardless of truth. At generation time the model samples or greedily selects high-probability tokens. Because fluency and factuality are governed by different things, a false claim can still be composed of individually high-probability tokens, yielding text that *reads* as confident. The sequence-level probability is a measure of linguistic typicality, not veracity, so a hallucinated answer can be assigned high likelihood. Methods such as semantic-entropy estimation show that meaningful uncertainty signal exists but requires sampling multiple generations and measuring *meaning*-level disagreement rather than reading off a single decoded string [2, 6].

(d) No clean internal "I know / I'm guessing" signal is surfaced by default. Probing work indicates models do carry *some* information about their own reliability—they can be trained to predict whether they know an answer [3]—but this signal is noisy, is not faithfully exposed in the free-text output, and is weakened by the confidence-rewarding post-training above. Verbalized confidence estimates are often poorly calibrated and must be elicited and corrected explicitly [7]. Consequently, in default use the model has no mechanism that converts internal uncertainty into an honest hedge; it emits the same assertive register whether the retrieved association is strong or spurious.

Synthesis. The model expresses confidence because confidence is cheap, stylistically rewarded, and statistically typical, while truth is never directly optimized and uncertainty is never reliably surfaced. Mitigations target exactly these points: calibration-aware training and evaluation that reward abstention [5], retrieval grounding, sampling-based uncertainty measures [2], and explicit confidence elicitation [7].

References

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